

STAT 4530/6530

Homework 1

Due: Friday, January 30 via eLC by 11:59 pm

Formatting instructions: Please label everything clearly and make sure that it is organized. I recommend typing your solutions, but neatly handwritten solutions are acceptable. For problems that require statistical computing, please attach relevant R code and plots. **Please write down (directly below your name) whether you are enrolled in STAT 4530 or STAT 6530.**

Grading: Homework accounts for 40% of your final grade, and all homework assignments contribute equally to your final homework score. Your score for each homework assignment will be recorded as a percentage of the total points that you received out of the total possible points. The amount of points possible for each problem are indicated in parentheses.

Problems

1. Consider the function $g(y)$ defined as

$$g(y) = \begin{cases} y^2, & \text{for } 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (i) (5 points) Is $g(y)$ a pdf?
- (ii) (5 points) If $g(y)$ is not a pdf, find the constant c so that $cg(y)$ is a pdf. Denote this pdf by $f(y)$.
- (iii) (5 points) Show that the cdf corresponding to the pdf $f(y)$ is given by $F(y)$,

$$F(y) = \begin{cases} 0, & \text{for } y \leq 0, \\ y^3, & \text{for } 0 < y < 1, \\ 1, & \text{elsewhere.} \end{cases}$$

- (iv) (5 points) For any $0 < p < 1$, find the inverse function $F^{-1}(p)$. Find the three quartiles of this distribution (in other words, find the 25th, 50th and 75th percentiles).
- (v) (5 points) Using the inverse cdf method, draw 10000 random samples from the distribution whose pdf is $f(y)$ and the cdf is $F(y)$. For this sample find the three sample quartiles (hint: the R function `quantile()` can be used to compute sample quantiles for a vector of samples). How do these quartiles compare with the values you found in part (iv)?
- (vi) (5 points) Using the 10000 observations that you have drawn in part (v), draw a relative frequency histogram. How does this histogram compare with the pdf $f(y)$? Comment by plotting the pdf $f(y)$ overlaid on the histogram.

2. Suppose Y is a continuous random variable with the pdf $f_Y(y)$ and the cdf $F_Y(y)$. For any number c ,

$$R(c) = \int_c^{\infty} f_Y(t) dt$$

is called the right-tail probability of Y at c . Note that $R(c) = 1 - F_Y(c)$. We can calculate $R(c)$ for any c . The function $R(Y)$ is a random variable (since Y is random).

- (a) (5 points) Find the cdf of $R(Y)$.
 - (b) (5 points) Find the pdf of $R(Y)$.
 - (c) (5 points) Find the mean and the variance of $R(Y)$.
3. Suppose Y is a standard normal random variable with cdf $\Phi(y) = P[Y \leq y] = \int_{-\infty}^y \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$.
- (a) (5 points) Show that for any c , $\Phi(c) + \Phi(-c) = 1$.
 - (b) (5 points) Let $X = \Phi(-Y)$. Find the cdf of X and the pdf of X .
 - (c) (5 points) Find the mean and the variance of X .
4. Consider two random variables X and Y .
- (a) (5 points) Show that $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$.
 - (b) (5 points) Show that $Var(X - Y) = Var(X) + Var(Y) - 2Cov(X, Y)$.
 - (c) (5 points) Show how (a) and (b) simplify when X and Y are uncorrelated.
5. Suppose a random variable Y has the exponential distribution with pdf $f(y; \lambda) = \lambda e^{-\lambda y}$ and cdf $F(y; \lambda) = 1 - e^{-\lambda y}$, for $y \geq 0$. Both the functions are zero for $y < 0$.
- (a) (5 points) Find the median of Y . Denote it by $\tilde{\mu}$.
 - (b) (5 points) Find the lower quartile and the upper quartile of Y .
 - (c) (5 points) How do the median and the upper quartile compare with the mean?
6. In an exit poll of 1648 voters in an election, 51.5% said they voted for Candidate A and 48.5% said they voted for Candidate B.
- (a) (5 points) Suppose that actually 50% of the population voted for Candidate A. If this exit poll had the properties of a simple random sample, find the standard error of the sample proportion voting for Candidate A.
 - (b) (10 points) Under the 50% presumption, are the results of the exit poll surprising? Why? Would you be willing to predict the election outcome? Explain by (i) conducting a simulation; (ii) using the value found in (a) for the standard error. (Hint: recall what you know about a normal distribution and how much of its density lies between two/three standard deviations of the mean.)

7. (10 points) Consider a sequence $X_1, X_2, \dots$ of independent $\text{Ber}(p)$ random variables, and let $T_n = \frac{1}{n} \sum_{i=1}^n X_i$. The mapping $g(x) = \log\left(\frac{x}{1-x}\right)$ is called the **logit** function and is used in statistics to map variables that are constrained to lie in the interval $(0, 1)$ to the real line. Use the delta method to find which distribution

$$\sqrt{n} \left[\log\left(\frac{T_n}{1-T_n}\right) - \log\left(\frac{p}{1-p}\right) \right]$$

converges in distribution to.

Solution: Because the variables $X_1, X_2, \dots$ are independent random variables that follow a distribution with mean equal to p and variance equal to $p(1-p)$ (or, equivalently, standard deviation equal to $\sqrt{p(1-p)}$), and because T_n is the sample mean, the central limit theorem implies that

$$\frac{T_n - p}{\sqrt{p(1-p)}/\sqrt{n}} = \frac{\sqrt{n}(T_n - p)}{\sqrt{p(1-p)}}$$

converges in distribution to a $N(0, 1)$ (standard normal) distribution. This is equivalent to the condition that $\sqrt{n}(T_n - p)$ converges in distribution to a $N(0, p(1-p))$ distribution. If this condition holds, some less mathematically precise (but common) ways to describe the situation are saying that “for a large sample size n , T_n is approximately normally distributed with mean equal to p and variance equal to $p(1-p)/n$ (or, equivalently, standard error equal to $\sqrt{p(1-p)}/\sqrt{n}$)” or “ T_n is asymptotically normally distributed with mean equal to p and variance equal to $p(1-p)/n$ (or, equivalently, standard error equal to $\sqrt{p(1-p)}/\sqrt{n}$).”

Because $\sqrt{n}(T_n - p)$ converges in distribution to a $N(0, p(1-p))$ distribution, applying the Delta method allows us to conclude that $\sqrt{n}(g(T_n) - g(p))$ converges in distribution to a $N(0, [g'(p)]^2 p(1-p))$ distribution, as long as $g(\cdot)$ is a smooth function. In this exercise, we have $g(x) = \log\left(\frac{x}{1-x}\right)$, so

$$g'(x) = \frac{1}{x(1-x)}$$

and

$$[g'(p)]^2 p(1-p) = \frac{1}{p^2(1-p)^2} * p(1-p) = \frac{1}{p(1-p)}.$$

It follows that

$$\sqrt{n} \left[\log\left(\frac{T_n}{1-T_n}\right) - \log\left(\frac{p}{1-p}\right) \right]$$

converges in distribution to a $N\left(0, \frac{1}{p(1-p)}\right)$ distribution. We could then say, in a less mathematically precise way, that “for a large sample size n , $\log\left(\frac{T_n}{1-T_n}\right)$ is approximately normally distributed with mean equal to $\log\left(\frac{p}{1-p}\right)$ and variance equal to $\frac{1}{np(1-p)}$.”

8. **This question is only required for students enrolled in STAT 6530.** (10 points) Chebyshev's inequality states that if X is a random variable with finite variance $\sigma^2 > 0$

and finite expected value μ , then for any positive real number k

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Use Chebyshev's inequality to prove the law of large numbers.